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Idle Well Utilization as Downhole Separation System to Improve Fuel Gas Quality: A Case Study in Jatibarang Field, Indonesia

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Abstract

PT Pertamina EP Regional 2 Zone 7 Jatibarang Field (PEPJTB) is one of the mature oilfields located in West Java, Indonesia. This field has been widely known as an oil producer for more than 50 years. Over the time, both production and reservoir pressure have declined. Currently, 900 barrels of oil per day (BOPD) are being produced and contributed by various artificial lift methods, one of them is Electrical Submersible Pump (ESP). Among these, ESP wells account for 51% of overall PEPJTB's total oil production.

During the operational phase, ESP systems in remote locations require generators positioned near the wellhead, with two available options: gas engine generators (GEG) and diesel engine generators (DEG). Due to gas treatment facility limitations, distance to well area and wet fuel gas sources, operational challenge appears such as repetitive well shutdowns and consequently, several ESP wells utilize DEG as the primary prime mover to maintain production with the associated cost arising.

PEPJTB personnel elaborated and developed an innovative solution to address the wet fuel gas issue. The team found out the space at certain depths of an idle well could potentially be used as further treatment to the fuel gas. A novel downhole separation method of fuel gas using idle wells has been investigated in this study. The downhole separator was installed alongside the completion string within the idle well and was placed above the fluid level. The fuel gas stream is injected into the tubing of the idle well and gravity-based separation occurs at the end of tubing (EOT) while the treated fuel gas flows upward through the annulus. The design referred to momentum principal of fluid flow when it passes the diverter or baffle plates attached inside the tubing. Based on the computational fluid dynamics simulation, the properties of injected fuel gas in terms of mass fraction of water and velocity have changed. The actual outcomes indicate favorable results since the moisture content of the fuel gas has decreased from 142 lb/MMscf to 39 lb/MMscf which is an acceptable level for the GEG. The treated fuel gas also allowed us to convert the ESP system from DEG to GEG, thereby avoiding high operational costs and meeting environmental compliance by reducing CO_2 emission equivalent by 1001 ton.

This paper aims to provide new insight into how the downhole separator has successfully improved ESP runtime and contributed to a reduction in loss of oil production opportunity (LPO), moreover, it has helped

in reducing CO_2 gas emissions in the remote area's ESP operation by retrofitting idle wells as a downhole separation system.

Introduction

Jatibarang field was found in 1968 which successfully produced oil and gas from Volcanic and Cibulakan Formation with peak production reaching 33,000 BOPD in 1974. After several years of production, the reservoir pressure has depleted therefore most wells can no longer flow naturally.

Various artificial lifts are being deployed to enhance the production, one of which is Electrical Submersible Pump (JTBESP). In operation mode, most of JTBESP use gas as fuel for the engine prime mover which the fuel gas is delivered from compressor station to well area. ESP may encounter challenges, from condensation of both water vapor and hydrocarbon in the fuel gas to limited gas treatment facilities in the remote area.

Case Study

The PEPJTB utilizes fuel gas to generate power for JTBESP operation. Gas is being compressed in main compressor station (MCS) and is delivered upon the gas network to the JTBESP well site. A simplified gas network system is illustrated in Fig. 1.

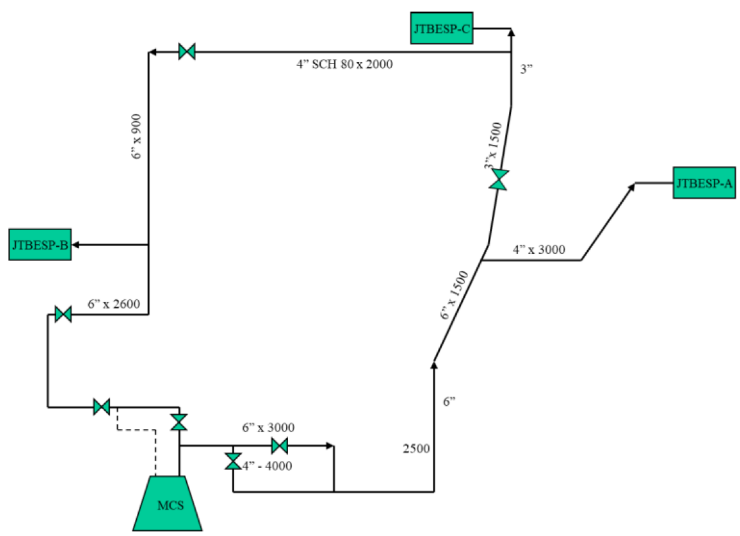

Figure 1—Simplified Gas Network System at Jatibarang Field

In addition, since JTBESP located in remote areas, it is powered by individual gas generator close to the wells with limited surface fuel gas treating facilities. The wellsite arrangement only consists of pressure regulator sets and a standalone main scrubber (Fig. 2). The distance between the well and discharge gas compressor station, as well as the wellsite configuration itself, can cause a drop in pressure and temperature, leading to condensation of both water vapor and hydrocarbon gas in the fuel gas (Gao et al., 2017).

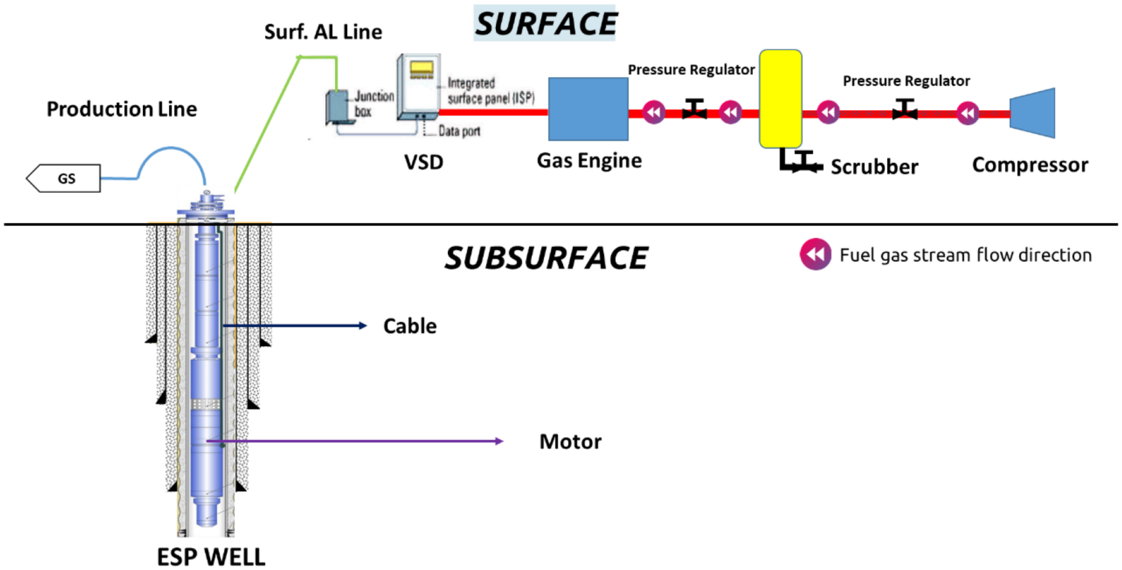

Figure 2—Typical JTBESP Wellsite Operational Arrangement

In 2023, several JTBESP units experienced frequent generator trips, causing sudden well shutdowns. This case study is represented by JTBESP-B, which produced at an average rate of 1,452 BLPD/20 BOPD. The data driven from downhole sensor of JTBESP-B is depicted in Fig.3.

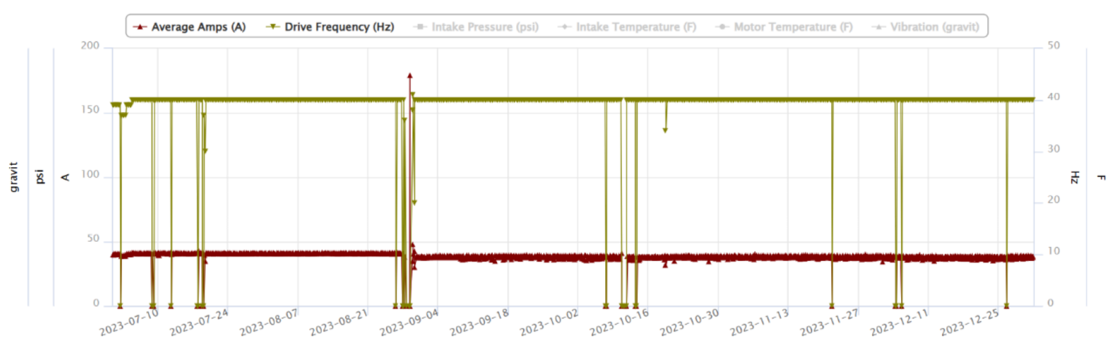

Figure 3—JTBESP-B Downhole Sensor

In July 2023, the downhole data sensor indicated that the well experienced a cumulative genset downtime of 12 hours, resulting in an LPO of 10 barrels of oil. This phenomenon led PEPJTB personnel to conduct several surveys, ultimately discovering the presence of liquid-water in the fuel gas stream by draining on-site scrubber.

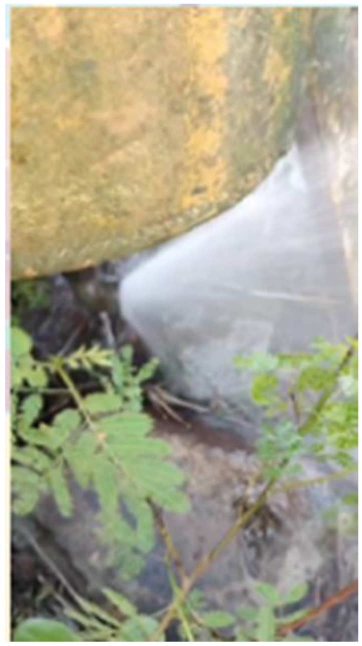

Figure 4—Illustration of Liquid-Water after On-Site Scrubber Drain (For Indicative Purposes Only)

A root cause analysis (Table 1) was performed to identify potential causes of liquid-water issues in the fuel gas. Several potential causes were identified, including an improper scrubber, fuel gas quality, ambient temperature, and limited operator allocation.

Table 1—Root Cause Analysis of Liquid-Water in the Fuel Gas Stream

Potential Causes	Detailed Cause and Effect	Likelihood
Improper Scrubber	The scrubber was undersized, so it couldn't effectively separate gas and liquid.	High
Quality of fuel gas	The fuel gas source for the gas engine may contain excess contaminants, such as water content	High
Ambient Temperature	Liquid-water condensing due to a drop in flow temperature	Medium
Limited Operator Allocation	Infrequent on-site bleed-off due to personnel limitation/other assigned duties	Low

From the analysis above, the main suspected causes were an improper scrubber and the quality of fuel gas as detailed below:

Improper Scrubber

As mentioned above, every JESP well pad is equipped with a set of pressure regulators and a main scrubber. The commonly installed main scrubber operates at approximately 250–260 psig, using a vessel size of 12 inches × 1.5 m. According to Svrcek and Monnery (1993), the recommended slenderness ratio (L/D) for a vertical separator or knock-out vessel operating at 250–500 psig is in the range of 3-4. Based on engineering calculations, the existing scrubber has L/D ratio of less than 2, making the separation process suboptimal.

Quality of fuel gas

Monitoring the quality of outflow gas from the well stream or surface equipment is a critical daily task. The gas sample was obtained from the flowline after the regulated stream. Laboratory results showed the gas had a moisture level of 142 lbs/MMscf, exceeding the acceptable limit for proper fuel gas quality.

Since most JTBESP units use gas as a primary mover fuel, it is essential to evaluate the quality of fuel gas and overall ESP system. The quickest possible solution to address this issue is by replacing the gas-fueled ESP engines with diesel-fueled engines. However, this would likely lead to higher operating costs and increased concerns about emissions. Various alternative fuel gas treatment options were identified, such as fuel gas separator and fuel gas filter. However, these solutions require relatively high investment costs. Moreover, installing equipment at the remote JTBESP-B cluster presents further complexities related to safety and security and does not achieve the LPO reduction target in a timely manner.

Given these circumstances, PEPJTB has identified and developed innovative solution using nearby idle well as a gas-liquid separation system. Gas-liquid separation is a technology used to separate the liquid phase from the gas phase through physical or chemical method. This paper attention is focused on how physical methods which commonly used by separator at surface facility equipment are implemented at downhole in the idle well configuration. The idle well selection must meet several requirements to accommodate the necessity of the downhole separation process.

Statement of Theory and Definitions

A separator, as illustrated in Fig. 5, is a mechanical apparatus designed to extract and collect liquids with varying specific gravities from natural gas streams. Sivalls (2009) explained that all separators are generally divided into four section or categories involving separation procedures: a) Primary separation section b), Secondary separation process, c) Liquid accumulation section and d) Mist extractor section.

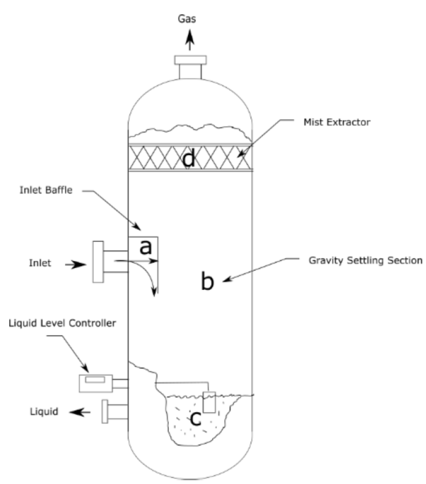

Figure 5—Gas-liquid separator schematic diagram (a) primary separation section; (b) secondary separation section; (c) liquid accumulation section; (d) mist extractor section.

Primary separation section

The primary separation section refers to the area surrounding the inlet within the separator internals, where the kinetic energy of the incoming fluid stream is effectively dissipated. This segment, along with its mechanical components, serves the essential function of initiating the separation of liquid from gas through various mechanisms such as deflectors or baffles. A significant portion of the liquid is directed toward the liquid collection zone, where larger volumes and droplets are rapidly separated due to gravitational forces.

In vertical separators, the inlet deflector redirects the liquid flow toward the vessel wall, forming a thin film that facilitates the release of dissolved gas. Conversely, in horizontal separators, the liquid typically encounters a deflector plate—either concave or flat—which guides it toward the vessel shell. This redirection helps isolate the liquid from the main gas stream and accelerates the liberation of solution gas. In certain configurations, baffles are utilized to fragment the liquid flow into finer streams and droplets, further enhancing the efficiency of gas separation.

Secondary separation section

The region of the separator located just beyond the inlet deflector—positioned between the liquid collection zone and the mist extractor—is referred to as the secondary separation section. In this area, the increased cross-sectional space reduces both gas velocity and liquid momentum, allowing liquid droplets to descend into the accumulation section under the influence of gravity.

In vertical separators, the upward gas flow opposes gravitational forces acting on the liquid particles, creating resistance that prevents smaller droplets from settling. Only particles with sufficient mass, in combination with gravitational pull, are able to reach the bottom. Finer droplets may remain suspended and exit the separator as entrained mist unless captured by a mist extractor.

In horizontal separators, the directional force of the gas stream acts perpendicular to gravity, which does not obstruct the downward movement of liquid particles. These droplets typically follow a diagonal trajectory toward the outlet. To ensure effective separation, horizontal separators must be adequately sized—both in cross-sectional area and length—to facilitate a reduction in gas velocity and provide sufficient residence time for liquid particles to settle into the collection zone.

Field studies have validated the optimal velocity parameters for the secondary separation section across various separator designs. Accordingly, sizing criteria are based on the permissible gas velocity within this section. Liquid particles that fail to settle are typically removed from the gas stream by the mist extractor integrated into the separator internals.

Liquid accumulation section

The liquid accumulation section within the separator internals comprises several key components: the liquid level controller, dump valves, control valves, and sight glasses. For two-phase separation, the typical liquid retention time is approximately one minute, which allows sufficient duration for solution gas to disengage from the collected liquid.

In vertical separators, a baffle plate is strategically placed between the liquid accumulation zone and the secondary separation section to aid in flow management. Horizontal separators, by contrast, generally allocate about half of this section for liquid collection. Due to their configuration, they possess lower surge capacity than vertical designs, and the retention time for three-phase separation is typically extended to around three minutes.

Regardless of orientation, liquid outlet connections are usually positioned as far from the inlet as possible to maximize retention time and promote effective gas release. These outlets are engineered with anti-vortex baffles or siphon-type drains to inhibit vortex formation, which can otherwise lead to gas re-entrainment in the discharged liquid.

Mist extractors

Mist extractors, integral components of separator internals, are typically categorized into three types commonly employed in oil and gas separation processes: knitted wire mesh, vane-type, and centrifugal units. Their primary function is to facilitate the coalescence of fine liquid particles into larger droplets. As the gas stream flows through the mist extractor, entrained liquid particles adhere to its surface, which becomes progressively wetted. Continued interaction with these wetted surfaces promotes the merging of smaller droplets into larger ones.

This process is further enhanced by the intricate and extended flow paths within the mist extractor, which alter the direction of the gas stream and increase contact time. Once the droplets attain a sufficient size to counteract the upward force exerted by gas velocity, they descend into the liquid accumulation section of the separator internals under the influence of gravity.

The Souders-Brown Approach

The Souders–Brown equation is widely utilized as a standard approach for determining the appropriate sizing of gas–liquid separators. According to Campbell (2015), when analyzing a spherical liquid droplet with a diameter denoted as D_P suspended within the gas phase, two primary forces influence its behavior, as illustrated in Fig. 6. The first is the drag force F_D, which arises from the motion of the gas and tends to carry the droplet along with the flow. The second is the gravitational force F_G, which results from the droplet's weight and acts to draw it downward, thereby facilitating its separation from the gas stream.

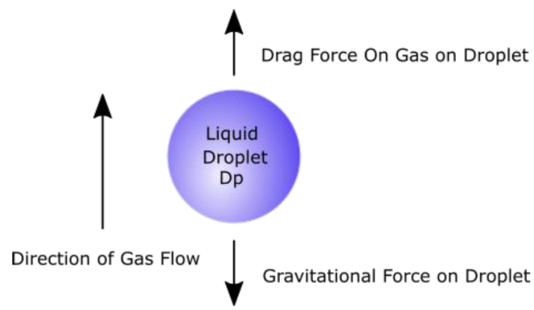

Figure 6—Schematic of the forces acting on a liquid droplet in the gas phase

Under the assumption of plug flow—characterized by the absence of eddies or disturbances—and considering a single droplet while neglecting end effects, equilibrium is achieved when the droplet reaches its terminal velocity. At this point, the drag force exerted by the gas flow is exactly balanced by the gravitational force acting on the droplet's mass.

$$F_D = F_G \tag{1}$$

By substituting the expressions for drag and gravitational forces into Eq. 1, the maximum permissible gas velocity, denoted as v_{Gmax}, can be determined. This velocity represents the threshold beyond which liquid droplets become entrained in the gas stream, thereby compromising separation efficiency.

$$v_{Gmax} = K_S\sqrt{\left(\frac{\rho_L - \rho_G}{\rho_G}\right)} \tag{2}$$

Eq. 2 is commonly known as the Souders–Brown equation, wherein K_S represents the design or sizing coefficient. The variables ρ_G and ρ_L correspond to the densities of the gas and liquid phases, respectively.

After determining the maximum allowable gas velocity v_{Gmax} using Eq. 2, the minimum required cross-sectional area of the separator vessel A_{min} can be calculated using Eq. 3. This ensures that the vessel is appropriately sized to prevent liquid entrainment and maintain effective phase separation.

$$A_{min} = \frac{V_g}{v_{Gmax}} \tag{3}$$

Where:

V_g = Gas volumetric flow rate

The design coefficient K_S in the Souders–Brown equation is an empirically derived parameter that plays a pivotal role in determining the appropriate diameter of gas–liquid separator vessels, as well as in sizing mist extractors. Its value is influenced by a range of operational and design variables, including:

- Operating pressure

- Physical properties of the fluids, with temperature exerting a significant effect on these characteristics
- Separator geometry and configuration
- Stability and consistency of flow conditions
- Design and efficiency of the inlet system
- Proportional flow rates of gas and liquid phases

A thorough evaluation of these factors is essential during the design process to ensure optimal separation performance and to prevent entrainment or carryover.

According to API Standard 12J, a recommended range of K_S values is provided for sizing vertical and horizontal gas–liquid separators. These values, which are outlined in Table 2, serve as design guidelines to ensure effective separation performance and minimize liquid carryover under varying operational conditions.

Table 2—API 12 J recommended range of K_S – values for vertical and horizontal separators

Type	Height or Length, ft (m)	Typical K_S range, ft/sec	Typical KS range, m/s
Vertical	5 (1.52)	0.12 to 0.24	0.037 – 0.073
	10 (3.05)	0.18 to 0.35	0.055 – 0.107
Horizontal	10 (3.05)	0.40 to 0.50	0.122 to 0.152
	Other Lenghts	0.40 to 0.50 $(L/10)^{0.56}$	0.122 to 0.152 $(L/3.05)^{0.56}$

The Idle Well Requirements

The idle well must meet the following requirements: a) It has high reservoir pressure to withstand the pressure of the injected gas that no gas losses to reservoir occur, b) Void which is left by cumulative production to collect the accumulated liquid, c) Close to the cluster where the fuel gas is needed and d) Good well integrity

Description and Application of Equipment and Process

The modified tubing and well configuration are illustrated in Fig. 7 with separation process as follow:

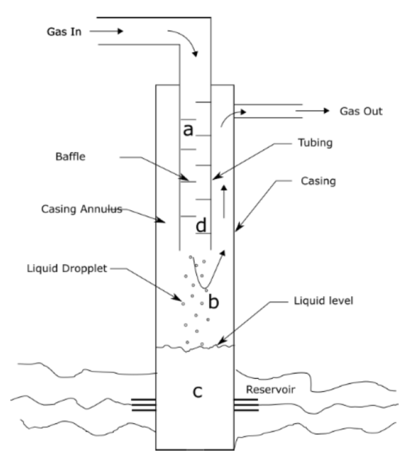

Figure 7—Downhole separator schematic diagram (a) primary separation section; (b) secondary separation section; (c) liquid accumulation section; (d) mist extractor section.

Primary separation section

Plates are installed to the body of tubing as deflectors or baffles where separation process by inertial impaction occur. Small liquid droplet formed at the surface of the baffles while gas proceed to the next baffles below and so on until reaching the end of the string.

Secondary separation section

This section is located between the end of tubing and surface where separation process by gravity occur. The liquid droplet will fall to the wellbore while the gas flows upwards through the casing annulus.

Liquid accumulation section

This section accommodated by the wellbore and void in reservoir which is left by cumulative production of the well in the past.

Mist Extractors

Mist extractors represented by the surface of the baffles where small droplets coalesce into bigger droplets which will increase the separation efficiency by gravity in the secondary separation section. Furthermore, the droplet from upper baffle will coalesce with droplet at lower baffle and so on.

Computational fluid dynamics simulation was carried out during the study as illustrated in Fig. 8. Based on simulation results, in terms of velocity profile, it can be observed the fluid velocity upon entry is approximately 80 m/s, and when it collides with the baffle, the fluid velocity has increased to 324.46 m/s due to the constriction of the flow area. Meanwhile, the water mass fraction has increased after the gas impacts the baffle plate.

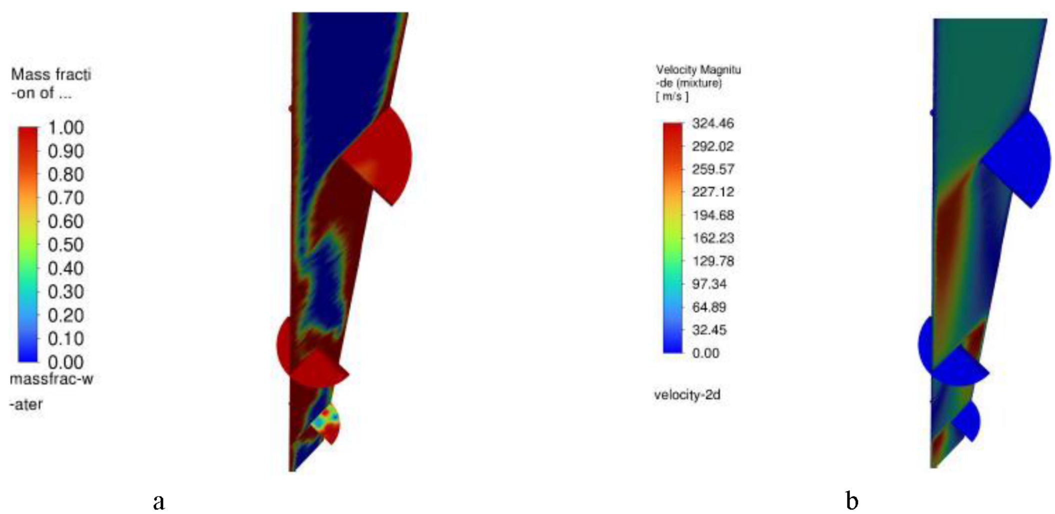

Figure 8—CFD simulation result. a) Mass fraction of water formed on the surface of baffle. b) Velocity of the gas stream increase after impact the baffle

The Idle Well Specification

The selected idle well's reservoir pressure is around 1849 psia with static fluid level at 400 m, while the pressure of the gas injected is 227 psia, enough to prevent gas losses to reservoir. The idle well has cumulative production 3 million barrels of oil which is more than enough to contain the volume of separated liquid. The idle well is located nearby the JTBESP-B well. Based on well intervention history, there is no issue regarding well integrity of the idle well. The production casing size is 9-5/8 in. and the tubing size is 2-7/8 in. One joint of modified tubing installed at 300 m depth along with standard tubing until surface. The rate of injected gas is 0.065 MMscfd.

Design Calculation

The data needed for design as follows:

$\rho_L = 63.05$ lb/ft^3

$\rho_G = 0.07$ lb/ft^3

$K_S = 0.35$ ft/s (Table 2)

$V_g = 0.065$ MMscfd

$ID_{casing} = 8.83$ in. (0.74 ft), $A_{casing} = 0.43$ ft^2

$OD_{tubing} = 2.875$ in. (0.24 ft), $A_{tubing} = 0.05$ ft^2

v_{Gmax} calculated using Eq. 2 gives 10.49 ft/s. Then calculate the minimum cross-sectional area using Eq. 3 which give A_{min} is 0.072 ft^2. Here the cross-sectional area represented by the area of the annulus between casing and tubing as illustrated in Fig. 9 which is 0.43 ft^2 subtracted by 0.05 ft^2 equals to 0.38 ft^2. This value is bigger than A_{min}, thus the configuration of the idle well is sufficient to provide effective separation.

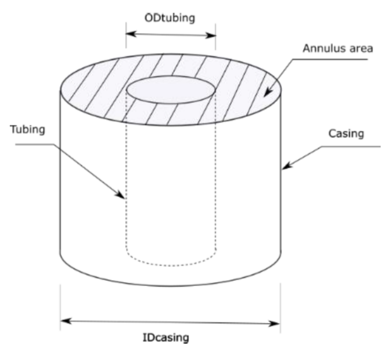

Figure 9—Cross-sectional area between casing and tubing as cross-sectional area of gravity settling section

Presentation of Data and Results

The implementation of downhole separator was scheduled within a seven-month timeframe, including monitoring and evaluation. The implementation steps are as follows:

1. Understanding the fuel gas problem through preliminary surveys and assessments.
2. Selecting a suitable well candidate.
3. Performing and analyzing pressure and temperature (P/T) surveys in selected pilot wells.
4. Conducting Simulation, Engineering Drawing and Modifying the tubing.
5. Executing the rig job in the selected idle well and setting the modified packer-less completion string.
6. Replacing the fuel gas supply line.
7. Converting DEG to GEG and Monitoring the new fuel gas system

JTBESP-B achieved notable results after implementing downhole separator in ESP operation. New ESP operation system illustrated in Fig 10. Based on the laboratory work, the moisture content from stream coming in and coming out of the idle well, there is reduction from 142 lb/mmscf to 39 lb/mmscf. After implementation the fluid level drop to 1043 m due to pressure of injected gas, while the perforation is at 2082 m. This indicate there is no gas losses to reservoir.

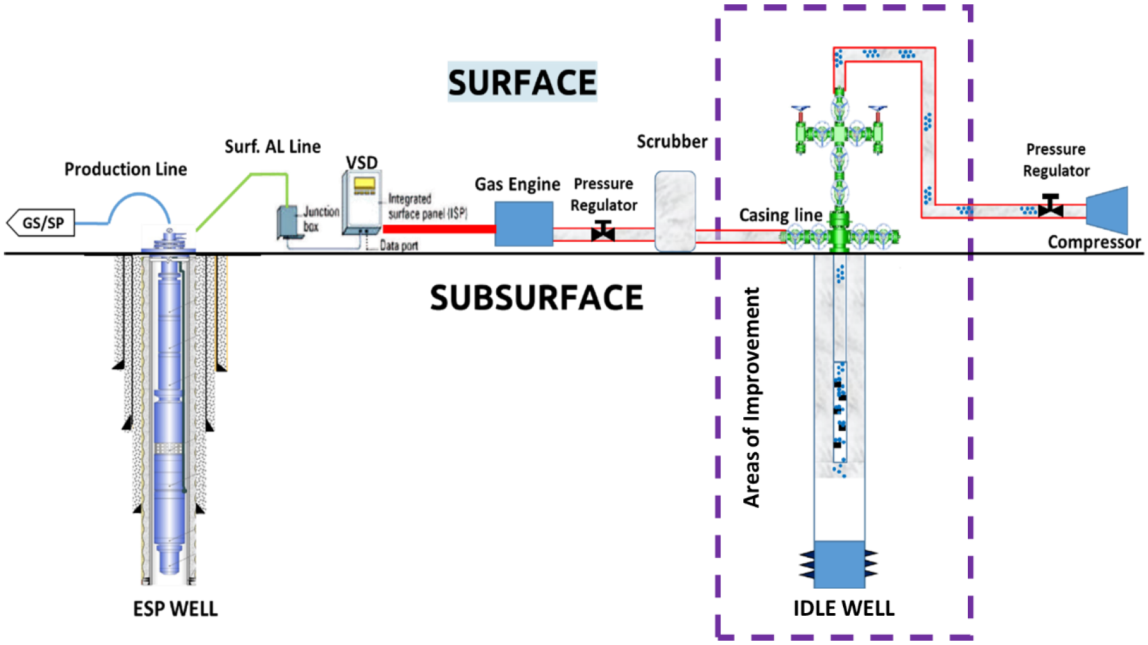

Figure 10—New ESP System Wellsite Arrangement

Upon downhole separator implementation as the new addition to fuel gas system, the treated fuel gas also allows the conversion of diesel-engine generator to gas-engine generator of ESP prime mover in the same well cluster (JTBESP-B cluster). This conversion lead to reduction of operating cost from using diesel to natural gas as fuel and reduction of CO_2e as well as the demonstrated outcomes in Table-3.

Table 3—Key Results

Parameter	Before	After
Moisture Content (lb/mmscf)	142	39
Operating Cost (USD/day)	1844	643
CO_2e (Ton/year)	1116	115
Production Loss (barrels oil/month)	10	0

Conclusion

Based on actual outcomes, configuration of modified tubing and utilization of idle well as downhole separator has proven to be an efficient solution for gas fuel with high moisture content problem at remote area and improved JTBESP-B's operation by minimizing its LPO.

The advantage of downhole separator compare to conventional surface separator as follows:

- It does not need extra space on surface at well cluster to provide more equipment for separation.
- The height of gravity settling section can be maximized to improve separation efficiency. In this case the height is 300 m, which is the distance between the end of tubing to the surface.

Additionally, PEPJTB has reported over 200 idle wells. This unique approach has provided valuable insights to unlock the opportunity by repurposing idle wells with sufficient well integrity as a downhole separation system. This method is both effective and environmentally friendly, particularly in a marginal field with facility limitations in remote well areas such as PEPJTB.

Future Steps

The geometry of baffle inside tubing can be altered to obtain better separation efficiency using CFD simulation study. Next study will be conduct to apply this idea to higher volume of gas such as gas well. The number of modified tubing can be added as needed based on CFD simulation study.

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Nomenclature

D_P	Diameter of liquid droplet
F_D	Drag force acting on the droplet due to gas flow
F_G	Gravitational force acting on the droplet
ρ_G	Density of the gas phase
ρ_L	Density of the liquid phase
v_{Gmax}	Maximum allowable gas velocity to prevent droplet entrainment
K_S	Souders-Brown design parameter, an empirical value used in separator sizing
A_{min}	Minimum cross-sectional area of the separator vessel
V_g	Volumetric flow rate of the gas phase
ID_{casing}	Inside diameter of Casing
OD_{tubing}	Inside diameter of Tubing
A_{casing}	Cross-sectional area of casing
A_{tubing}	Cross-sectional area of tubing

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